

Koussay Akchi

Cloud Engineer

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Profile

Full Stack & Cloud Engineer passionate about scalable and robust web architectures, cloud-native systems, and DevOps automation. Experienced across modern backend and frontend stacks (React, Node.js, Laravel, Python) and cloud infrastructure (Azure, Kubernetes, Docker, Terraform, CI/CD). Actively building and sharing production-grade, open-source projects.

Experience

Cloud Engineering Intern

Jun 2026 - Aug 2026

Next Step | Tunisia

Designed and deployed a unified API and AI Gateway platform on Apache APISIX, centralizing API management, load balancing, and AI orchestration in a Kubernetes-native environment.

- Built AI gateway orchestration with multi-model routing (ai-proxy-multi), prompt templating and decoration, dynamic request rewrites, and PII masking / data anonymization guardrails.
- Configured edge traffic management using distributed rate/connection limiting, traffic splitting, circuit breaking (api-breaker), IP/consumer restrictions, and request mirroring.
- Implemented payload transformations, gRPC transcoding, response rewrites, and schema validation with custom Lua and serverless functions.
- Streamlined observability and centralized audit streaming via OpenTelemetry, Prometheus metrics, Kafka, and Elasticsearch logging pipelines.
- Delivered a React and Node.js self-service portal with RBAC and monitoring dashboards.
- Validated service resilience using chaos engineering and automated CI/CD deployment pipelines across cloud-native environments.

Tools: Apache APISIX, Kubernetes, Helm, Lua, Kafka, Prometheus, Grafana, Jaeger, Redis, Node.js, React, PostgreSQL

Full Stack Software Engineer

Jul 2025 - Present

OpkodeLabs | Tunisia

Developed WattLink, a multi-tenant EV charging management platform built on the OCPP 1.6 protocol, with real-time visibility into station status, pricing, sessions, and telemetry data.

- Built a real-time monitoring system using Laravel, PostgreSQL, Redis Pub/Sub, Socket.IO, and Grafana.
- Implemented remote operator controls for charging sessions, pricing updates, and live telemetry dashboards.
- Designed secure multi-tenant workflows to isolate data and operations across customer networks.
- Delivered production-ready infrastructure and tooling with Python, Docker, and reliable service orchestration.

Tools: Laravel, Python, Redis, Docker, Socket.IO, Grafana, PostgreSQL

Full Stack Software Engineer

Feb 2025 - Dec 2025

HexaFlow | Tunisia

Developed dynamic web applications with React, Next.js, Node.js, and Java while improving system architecture with containerized deployments and cloud-ready patterns.

- Delivered customer-facing features in agile teams with a focus on maintainability and timely releases.
- Improved deployment workflows by integrating Docker and cloud infrastructure tooling.
- Enhanced frontend performance, state management, and component modularity for seamless user experiences.
- Integrated automated CI/CD checks and containerized environments to ensure reliable multi-service deployments.

Tools: React, Next.js, Node.js, Java, Docker, Terraform, Azure

Full Stack Software Engineer

Jun 2024 - Jan 2025

Wecraft | Tunisia

Developed and maintained a CRM platform for pharmaceutical laboratories, covering both the user interface and backend business logic for research, sales, and compliance workflows.

- Implemented front-end interfaces for laboratory operations and internal data entry flows.
- Built backend services and business rules to support product, customer, and order management.
- Added automated testing across the stack to reduce regressions and improve release stability.
- Engineered robust RESTful endpoints with role-based access control and token lifecycle verification.

Tools: React, Quarkus, Docker, JWT, PostgreSQL, MongoDB

End of Studies Internship

Jan 2024 - Jun 2024

Wecraft | Tunisia

Worked on the CRM for pharmaceutical laboratories during my internship, contributing across the stack while learning production-grade web app architecture and development workflows.

- Participated in the design and initial implementation of the application architecture.
- Contributed to both front-end features and back-end services for core business processes.
- Collaborated with the team to improve code quality, maintainability, and deployment practices.
- Structured data schemas and storage strategies across relational and NoSQL databases for laboratory workflows.

Tools: React, Quarkus, Docker, JWT, PostgreSQL, MongoDB

Summer Internship

Jun 2023 - Jul 2023

Alliance Technologie | Tunisia

Developed a microservice and API for managing electronic purchase vouchers for SNDP customers.

- Built backend logic for voucher lifecycle management and validation flows.
- Integrated application services with secure data handling and business rules.
- Collaborated on API design and service integration to support customer workflows.
- Automated database migrations and transactional queries ensuring strict financial integrity for transactions.

Tools: C#, ASP.NET, Java, PostgreSQL

Education

Engineering Degree in Software Engineering

2024 - Present

ESPRIT

Bachelor's Degree in Computer Science (GLSI)

2021 - 2024

ISTIC Borj Cedria

Skills

Frontend: Next.js, React, Angular, Flutter, HTML, CSS

Backend: Node.js, Laravel, Spring, Quarkus, MongoDB, PostgreSQL, Redis, REST APIs, GraphQL

Programming Languages: Java, Python, TypeScript, JavaScript, PHP, C, Lua

Cloud / DevOps: Terraform, IaaS (Infrastructure as Code), CI/CD, Jenkins, Azure, Kubernetes, Docker, Ansible, Linux, OpenStack, Grafana, Prometheus, Helm, Apache APISIX, Kafka, Jaeger, OpenTelemetry

AI: PyTorch, TensorFlow, Hugging Face, Ollama, OpenAI, LLMs, AI/ML workflows

Other Tools: Figma, MediaWiki, Postman, PostHog, SonarQube, Vercel, Firebase

Languages

English: Fluent • French: Fluent • Arabic: Native

Certifications

- Microsoft Applied Skills: Deploy cloud-native apps using Azure Container Apps - Microsoft (Sep 2026)
<https://learn.microsoft.com/api/credentials/share/en-us/koussayakchi/9FC89CD5F97D781C?sharingId=7C446232C661FA23>
- Microsoft Applied Skills: Configure secure access to your workloads using Azure networking - Microsoft (Sep 2026)
<https://learn.microsoft.com/api/credentials/share/en-us/koussayakchi/FAE4428DA68E605F>
- Microsoft Applied Skills: Get started with cloud security and monitoring tasks - Microsoft (Sep 2026)
<https://learn.microsoft.com/en-us/users/koussayakchi/credentials/6df9c9e21c5b253>
- Microsoft Applied Skills: Get started with Azure management tasks - Microsoft (Sep 2026)
<https://learn.microsoft.com/api/credentials/share/en-us/koussayakchi/63B2C35CA975068F>
- CCNA: Switching, Routing, and Wireless Essentials - Cisco (Jul 2025)
<https://www.credly.com/badges/058d5f66-db69-484d-88be-525bb33c98c9>

Projects

Cluverse - Cloud-Native AI-Centric Club Management Platform

[Source Code \(GitHub\)](#)

Engineered an enterprise-grade, microservice-based SaaS platform modernizing student club operations with private cloud infrastructure, self-healing orchestration, and applied AI intelligence.

- Automated private cloud infrastructure on OpenStack using Heat templates and bootstrapped a multi-node Kubernetes cluster (v1.35.1) with Ingress-NGINX, Spring Cloud Gateway, and Cloudflare Tunnel.
- Engineered multi-layer cluster observability and predictive AI telemetry combining Prometheus, Alertmanager, Grafana, and an embedded Meta Prophet time-series engine for preemptive resource anomaly detection.
- Integrated Spring Boot 3.4 microservices with autonomous AI services: FastAPI interview speech analyzer (Whisper v2 + Groq Llama 3.1), cashflow forecasting model, and Stripe payment fraud detection.
- Implemented automated configuration management and zero-downtime rolling updates via Ansible playbooks and container image polling daemons.

Tools: OpenStack, Kubernetes, Spring Boot, FastAPI, Python, Meta Prophet, Ansible, Prometheus, Grafana, Docker, MySQL, Redis

Shorten - Production URL Shortener & Analytics

[Demo / Website](#) | [Source Code \(GitHub\)](#)

Engineered a production-ready, high-throughput URL shortening service featuring custom vanity slugs, automated link lifecycle management, and real-time per-link analytics.

- Implemented high-performance redirection layer with custom slug hashing, collision detection, and TTL/expiry logic on MongoDB Atlas.
- Integrated privacy-friendly per-link click tracking, user-agent analytics, and cookie-based persistent session history without user friction.
- Built a multi-stage GitHub Actions CI/CD pipeline enforcing automated ESLint code quality gates, SonarCloud static analysis, and multi-arch Docker image packaging.

Tools: Next.js, React, TypeScript, MongoDB Atlas, Docker, GitHub Actions, SonarCloud

TirFly - AI-Powered Travel Management Platform

[Source Code \(GitHub\)](#)

Architected an end-to-end travel platform combining agency back-office operations with client booking workflows for flights, accommodations, customized vacation packages, and events.

- Integrated machine learning models for personalized travel recommendation engines and dynamic sentiment/feedback analysis.

- Developed robust integrations with third-party travel APIs for live flight schedules, hotel inventory, and real-time exchange rates.
- Implemented role-based access control (RBAC) separating administrative CRUD workflows from customer self-service reservation lifecycles.

Tools: Symfony, PHP, Bootstrap, Tailwind CSS, REST APIs, Python / AI Models, MySQL

AI Vehicle Tracking & Automated License Plate Recognition (ALPR/ANPR)

[Source Code \(GitHub\)](#)

Developed a high-throughput computer vision pipeline for multi-object vehicle detection, trajectory tracking, and automated license plate character recognition from real-time video streams.

- Built real-time vehicle detection using YOLO object detection models with optimized bounding-box localization.
- Designed continuous multi-vehicle tracking using Structural Similarity Index Measure (SSIM) and temporal frame matching to maintain distinct identity across occlusions.
- Engineered contour analysis heuristics and thresholding algorithms to extract license plate regions of interest, feeding segmented character streams into Tesseract OCR for text extraction.

Tools: Python, OpenCV, YOLO, SSIM, Tesseract OCR, NumPy

Sound-Grid - Interactive Web Audio Synthesizer & Sequencer

[Demo / Website](#) | [Source Code \(GitHub\)](#)

Created an interactive browser-based step sequencer and audio visualizer for polyphonic loop composition.

- Engineered low-latency audio scheduling and polyphonic playback using the native HTML5 Web Audio API.
- Designed a responsive 60fps grid matrix interface with state synchronization and custom tempo/BPM clock synchronization.

Tools: JavaScript, Web Audio API, HTML5 Canvas / CSS3, State Management

FTL AutoSave Manager - Desktop Save State & Backup System

[Source Code \(GitHub\)](#)

Engineered a cross-platform desktop automation utility for real-time game save state tracking, versioning, and rollback management for Faster Than Light (FTL).

- Implemented automated filesystem watchers and event hooks to detect file modification events and trigger atomic snapshot backups.
- Engineered save state branching, metadata serialization, and one-click restoration workflows via a responsive Java GUI.

Tools: Java, Swing / JavaFX, Filesystem I/O, Concurrency / Threading

Grid Image Viewer - High-Speed 2x2 Image Triage Utility

[Source Code \(GitHub\)](#)

Built an optimized desktop media triage utility designed for rapid categorization, tagging, and organization of large local image datasets.

- Constructed a multi-threaded thumbnail loader utilizing memory caching and fast image decoders to prevent UI freezes during large batch browsing.
- Integrated hotkey-driven atomic file operations (copy, delete, move) with non-destructive undo buffers and session statistics tracking.

Tools: Python, Tkinter, Pillow (PIL), Multithreading, OS / Filesystem

Minesweeper - State-Driven Game Engine Clone

[Demo / Website](#) | [Source Code \(GitHub\)](#)

Implemented a full-featured browser recreation of classic Minesweeper emphasizing deterministic board generation and algorithmic state resolution.

- Implemented recursive flood-fill algorithms for zero-cell reveal cascade operations with optimal computational complexity.

- Engineered customizable grid dimensions, difficulty parameters, mine-density configurations, and high-precision timer tracking.

Tools: React, JavaScript (ES6+), CSS3 / CSS Grid, Algorithms